

drive system, e.g., a linear servo motor. You may also require to count an event which is signaled by a voltage pulse, such as in flywheel counters or robotic shaft encoders.

Most of the commercially available DAQ systems come with all the above-mentioned features of counters, analogue inputs and outputs, digital inputs and outputs, etc. However, these systems are quite expensive and come with a paid driver for interfacing. Most of them can only be used with the proprietary software of the manufacturer.

This project is intended for those who want to make their own system for data acquisition due to the limitation of their total project cost or installation where one cannot use commercially available DAQ (such as in a nuclear or defense installation). The author's prototype is shown in Fig. 1.

Circuit and working

Fig. 2 shows the circuit of USB data acquisition system.

AT90USB1287. The heart of the system is a 64-pin ATMELE USB chip AT90USB1287, which is an 8-bit microcontroller that can do USB communication and take analogue input through its eight 10-bit analogue-to-digital converters (ADCs). It has 48 programmable input/output pins. The microcontroller runs on a 16MHz clock and has 128 kB of Flash memory. As the microcontroller does not have any digital-to-analogue converter (DAC) on its output pins, we required an external chip for this purpose. We used TLC7226 from Texas Instruments (TI), which is an 8-bit DAC with quad outputs.

Before finalising the design, ensure that all the AVR hardware design considerations mentioned in ATMELE application note are met. You can easily get IC samples for academic and research purposes, which the manufacturers always support.

TLC7226. This is a 20-pin dual-inline chip. You may refer to its datasheet from TI for more details. This IC is driven by a reference diode LM336

PARTS LIST

Semiconductors:	
IC1	- AT90USB1287 microcontroller
IC2	- TLC7226 quad DAC
IC3, IC4	- OPA2335 op-amp
IC5	- LM336 programmable shunt regulator
LED1, LED2	- 5mm LED
Resistors (all 1/4-watt, $\pm 5\%$ carbon):	
R1-R8	- 47-kilo-ohm
R9	- 1-kilo-ohm
R10	- 1.5-kilo-ohm
R11, R15	- 4.7-kilo-ohm
R12	- 470-ohm
R13, R14	- 220-ohm
R16	- 680-ohm
Capacitors:	
C1-C4, C6, C7	- 100nF ceramic disk
C5	- 1 μ F, 16V electrolytic
C8, C9	- 22pF ceramic disk
C10	- 0.47 μ F ceramic disk
Miscellaneous:	
L1	- 10 μ H inductor
X _{cr} , 1	- 16MHz crystal oscillator
SI-53	- Tactile switch
CON1, CON2	- 18-pin bergstrip male connector
CON3	- USB type-B connector
CON4	- 2-pin bergstrip connector
CON5	- 6-pin ISP/ICSP male connector
SJ1	- Shorting jumper connector

to have a regulated voltage supply of 2.5V. So for an input ranging from 0 to 255 through parallel eight pins of the microcontroller, the output varies from 0 to 2.5V. As the output desired is 5V, we have used a voltage amplifier with gain set at '2.' For voltage amplification we used quad op-amp OPA2335 from Texas Instruments, which has zero drift and low offset voltage of 5 μ V.

Power supply. The circuit is powered from the USB port, which provides a voltage output of up to 4.5V. If you need to connect the output pins to a load larger than that supported by your USB port (typically 500 mA per port), use an external power source of 5V and remove the USB power jumper (SJ1) shown in Fig. 6.

Connectors. Since the system requires interfacing with an external device for input and output, we have used connectors for various purposes. One 18-pin connector (CON2) is used for analogue inputs and outputs, while another 18-pin connector (CON1) is used for digital inputs and outputs. There is also a 6-pin connector for ISP interfacing and a 2-pin shorting jumper

connector for power supply from the USB port.

Specifications of the developed DAQ device are listed in Table I.

Construction

Once done with the schematic of the project, you need to mount the components on the provided PCB. An actual-size, double-side, solder-side PCB track layout of USB data acquisition system is shown in Fig. 3 and component-side track layout in Fig. 4. The component layout is shown in Fig. 5. The author's assembled board with input and output pin details is shown in Fig. 6.

The original board routing and layout design was done by the author on a four-layered board with inner layers supplying the power. Having continuous copper layers in the inner layers provided better stability to ADC and DAC systems.

The board was built with a mix of surface-mount device (SMD) and through-hole technology (THT) components as we could not obtain small quantities of the discrete components in SMD packages. You can choose to use only SMD components to have a compact design. You can also try assembling the circuit on a double-layered board to cut down the development cost.

Firmware and software design

The firmware was developed using open source library 'Lightweight USB Framework for AVR,' commonly known as LUFA. It requires free compiler WinAVR GCC 2010 to be installed in your PC. It is advised to install AVR Studio 4 before WinAVR as programming and compilation are easier in AVR Studio. Now load project file GenericHID.aps file (from www.efymag.com or EFY DVD of this month's EFY Plus) into AVR Studio environment. Compile the code to generate the hex code.

Burning the hex code. Microcontroller AT90USB1287 comes with a preloaded bootloader program that can